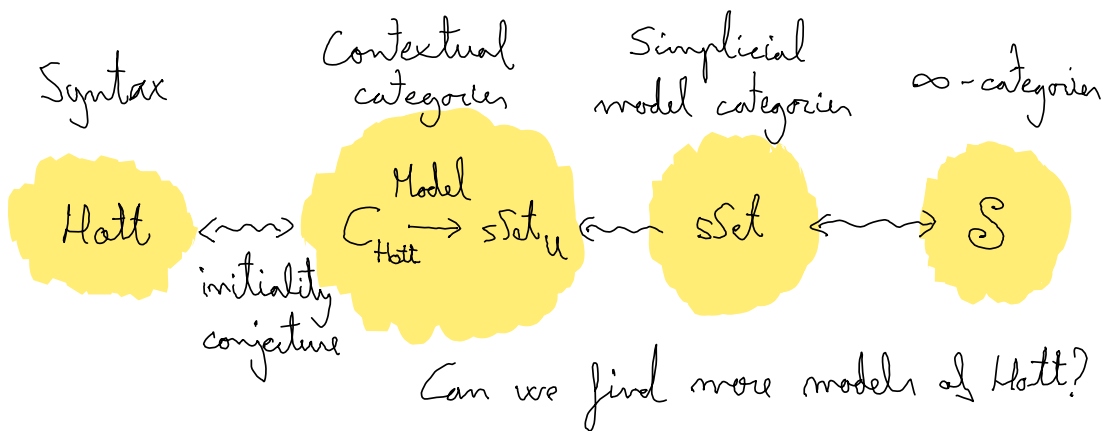


On the ∞ -topos semantics of HoTT

Part 1: Models of ∞ -toposes and proof sketch

References:

- [Riehl 2022, On the ∞ -topos semantics of HoTT]
Main reference, this talk follows lecture 3
- [Shulman 2019, All ∞ -toposes have strict...]
Original proof of ∞ -topos semantics
- [Lumbraine & Warren 2015, Local universes model]
[Aroldy 2018, Natural models of HoTT]
Solution for the coherence problem



1. ∞ -categories via simplicial model categories

∞ -category $\approx$ category "weakly" enriched in
" ∞ -groupoids" ($\approx$ spaces)

Thm (Bergner, Lurie) Any ∞ -category can be
presented by a simplicially enriched category

∞ -categories $\longleftrightarrow$ Kan complex enriched
categories

$\Rightarrow$ Any Kan complex enrich category presents
an ∞ -category up to equivalence of ∞ -cat.

Main example: simplicial model categories

Ex. $s\mathbf{Set}$ is a simplicial model category,
and presents $\mathcal{S}$, the ∞ -category of spaces.

A complete and cocomplete category C is a model category if it has three collections of morphisms $Weg(\xrightarrow{\sim})$, $Fib(\twoheadrightarrow)$ and $Cof(\twoheadrightarrow)$ such that

(1) Weg satisfy 2-out-of-3

(2) $(Cof \cap Weg, Fib)$ and $(Cof, Fib \cap Weg)$ are weak factorisation systems

In addition, C is simplicial if it is simplicially enriched:

$$Map: C^{op} \times C \rightarrow sSet$$

and for all $A \xrightarrow{i} B$ and $X \xrightarrow{j} Y$

$$Map(B, X) \longrightarrow Map(A, X) \times_{Map(A, Y)} Map(B, Y)$$

is a right Quillen functor.

$\Rightarrow$ If C is a simplicial model category, the fibrant and cofibrant objects define a Kan complex enriched category $\rightsquigarrow \infty$ -category!

2. ∞ -toposes via model toposes

Goal: generalise the model of Hott for $\mathcal{S}$ to:

- Diagrams over $\mathcal{S} \rightsquigarrow \mathbf{PSh}(A) = \mathbf{Fun}(A^{\mathrm{op}}, \mathcal{S})$
- Localisation of diagrams over $\mathcal{S} \rightsquigarrow \infty\text{-topos}$

(Similar to: $\mathbf{Set} \rightsquigarrow \mathbf{PSh}(A) \rightsquigarrow \mathbf{Sh}(A, \mathbb{Z})$
Grothendieck A -topos)

Given a model cat M and a simplicially enriched small category A , can we give a model structure to $\mathbf{Fun}(A^{\mathrm{op}}, M)$?

Idea. $\forall X, Y \in \mathbf{Fun}(A^{\mathrm{op}}, M)$,

$$\begin{array}{ccc} f: X \rightarrow Y & & f_a: X_a \rightarrow Y_a \\ \text{Weg/Fib/Cof in } \mathbf{Fun}(A^{\mathrm{op}}, M) & \iff & \text{Weg/Fib/Cof in } M \\ & & \forall a \in A \end{array}$$

Problem. Breaks the weak factorization systems

Solution. The injective model structure on $\text{Fun}(A^{\text{op}}, M)$ has for every $f: X \rightarrow Y$:

- $f \in \text{Weg} \Leftrightarrow f_a: X_a \rightarrow Y_a \text{ Weg of } M \forall a \in A$
- $f \in \text{Cof} \Leftrightarrow f_a: X_a \rightarrow Y_a \text{ Cof of } M \forall a \in A$
- $f \in \text{Fib} \Leftrightarrow f \text{ has RLP with } \text{Cof} \cap \text{Weg}.$

$\leadsto$ We are fixing Weg and Cof , and taking Fib such that we have the weak fact. systems.

$\leadsto$ There is a projective model structure, which has Weg and Fib pointwise.

$\leadsto$ Both are Quillen equivalent via identities.

Ex. $\text{PSh}(A) = \text{Fun}(A^{\text{op}}, \text{sSet})$

The injective model structure $\text{PSh}(A)_{\text{inj}}$ is a simplicial model category, which presents the ∞ -category of presheaves over the small ∞ -category presented by A .

Localization of a model structure is the process which "formally inverts" the maps on a set by turning them into weak equivalences.

A **left Bousfield localization** LM of a model category M is another model structure on M with the same cofibrations and more weak equivalences.

Thm. If M is a left proper, combinatorial, simplicial model category and $S \subset \text{Mor}(M)$ is a set of morphisms, then $L_S M$ exists and is left proper, combinatorial and simplicial.

Ex. For any set $S \subset \text{Mor}(\text{PSh}(A))$, $L_S(\text{PSh}(A)_{\text{inj}})$ presents a presentable ∞ -category.

$\leadsto$ Any presentable ∞ -category arises as a left Bousfield localization of $\text{PSh}(A)_{\text{inj}}$.

A **model topos** $\mathcal{E}$ is a simplicial model category Quillen equivalent to a left Bousfield localization of $\mathbf{PSh}(A)_{\text{inj}}$ such that

$$L_S(\mathbf{PSh}(A)_{\text{inj}}) \begin{array}{c} \xleftarrow{L} \\ \xrightarrow{R} \end{array} \mathbf{PSh}(A)_{\text{inj}}$$

where L preserves finite homotopy limits.

An **∞ -topos** $\mathcal{E}$ is an accessible and left exact localization of the ∞ -category of presheaves over some small ∞ -category, i.e.,

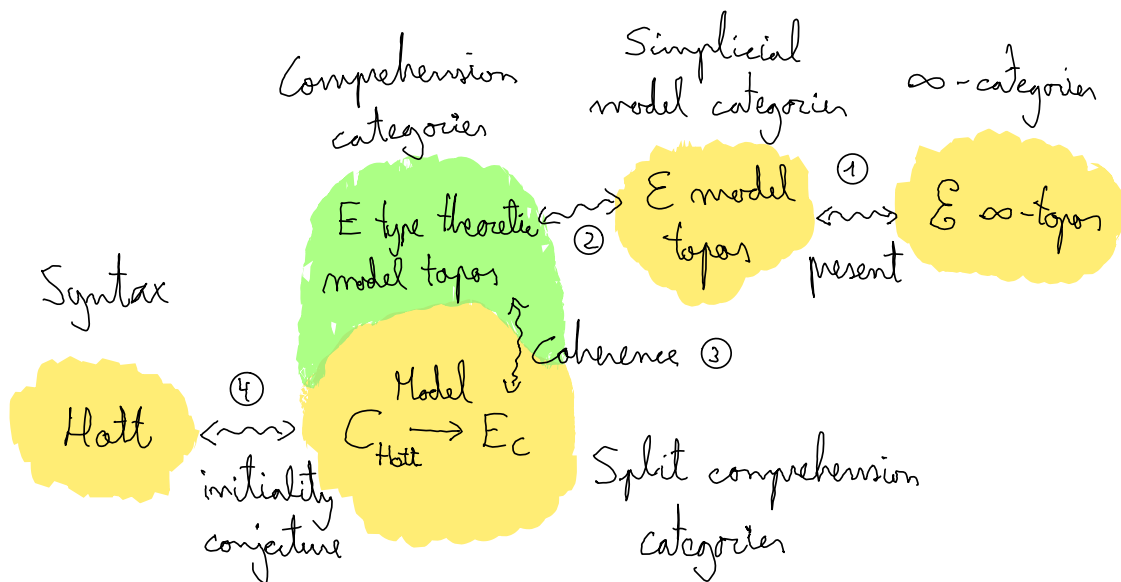
$$\mathcal{E} \begin{array}{c} \xleftarrow{L} \\ \xrightarrow[\perp]{R} \end{array} \mathbf{PSh}(A) = \mathbf{Fun}(A^{\text{op}}, \mathcal{S})$$

where R is accessible and fully faithful and L preserves finite limits.

Ex. Any ∞ -topos is presented by a model topos, and any model topos presents an ∞ -topos.

3. ∞ -topos semantics of Hott

- A model cat C is a type theoretic model topos if:
- (i) its underlying category is a Grothendieck 1-topos
 - (ii) it's a simplicial, combinatorial, right proper model structure with monos as cofibrations,
 - (iii) it's simplicially locally cartesian closed
it has a locally representable and relatively
acyclic notion of fibant structure



- ① Any ∞ -topos is presented by a model topos, and any model topos presents an ∞ -topos
- ② Model topos $\iff$ Type theoretic model topos
- ③ The 2-category of comprehension categories and split comprehension categories are equivalent:
For any type theoretic model topos, there is an equivalent "strict" model of Hott.
[Appendix A of Shulman, Lumsdaine-Warren, or Awodey]
- ④ Initiality conjecture: the comprehension category of Hott is the initial object in the category of comprehension categories.

Next day: prove ②